

Alzheimer's HybridRAG

GraphRAG vs VectorRAG vs Hybrid

Generated: 2026-07-24 19:14 UTC

Generator: google/medgemma-4b-it

Knowledge graph: 1039 nodes / 1356 edges

Vector index: 1488 chunks (NeuML/pubmedbert-base-embeddings)

Questions: 54 Alzheimer's MIRAGE items

Strategies: closed_book, hybridrag, hybridrag_gated

Self-consistency: off (greedy)

Best: closed_book = 0.6481

Baseline: 0.6481

Runtime: 29 min

Results

strategy	acc	95% CI	fixed	broke	p
closed_book	0.6481	[0.515,0.762]	-	-	base
hybridrag	0.6296	[0.496,0.746]	7	8	1.0000
hybridrag_gated	0.6481	[0.515,0.762]	1	1	0.4795

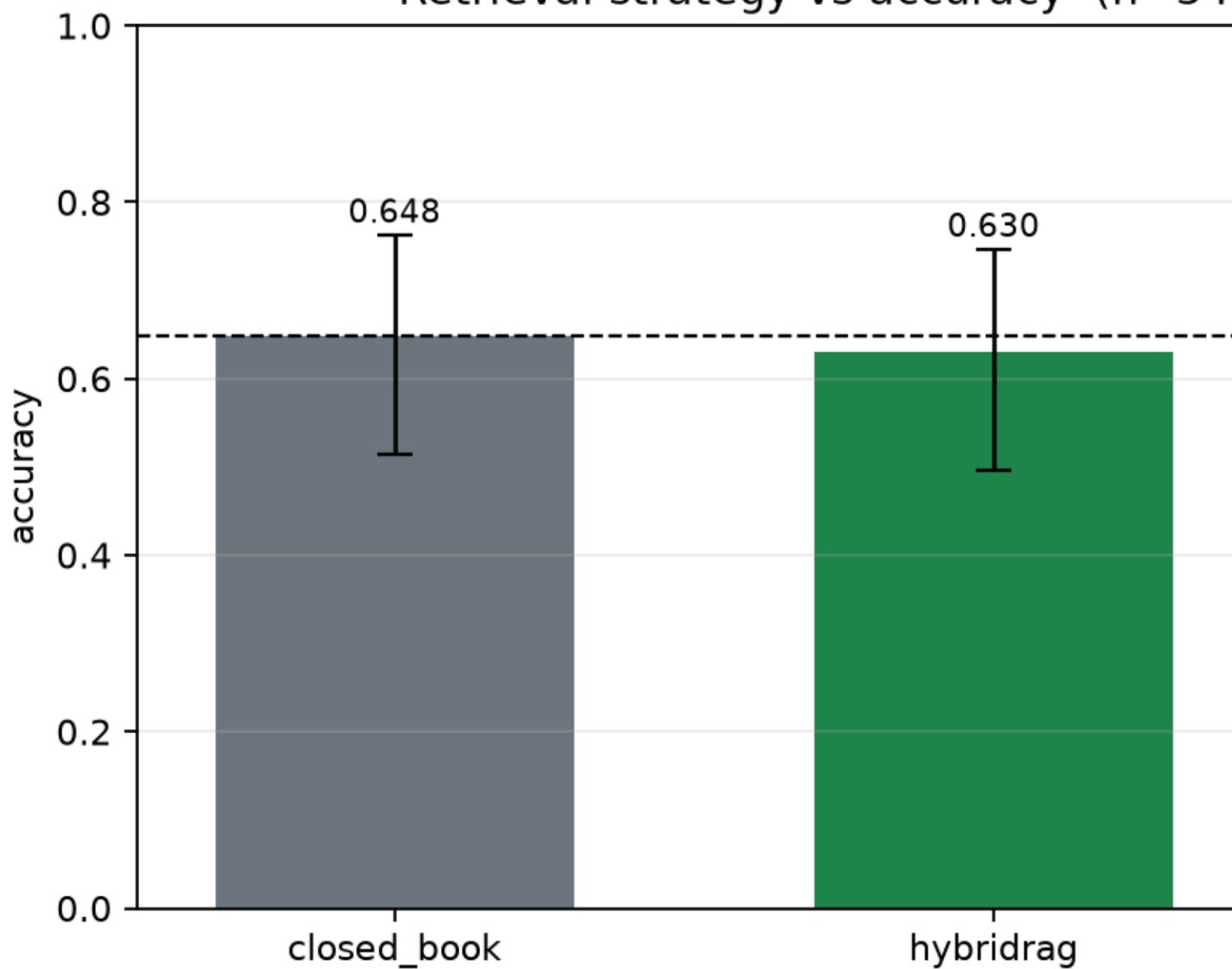
head-to-head:
hybridrag_vs_hybridrag_gated fixed 7 broke 6 p=1.0000

retrieval coverage:
KG entity-linked 52/54
KG bridge facts 14/54
vector mean top-1 cosine 0.531
mean closed-book margin 0.9257

evidence volume (mean per question):
closed_book facts 0.0 chunks 0.0 ctx 0 chars
hybridrag facts 11.6 chunks 10.0 ctx 5941 chars
hybridrag_gated facts 0.8 chunks 0.6 ctx 296 chars

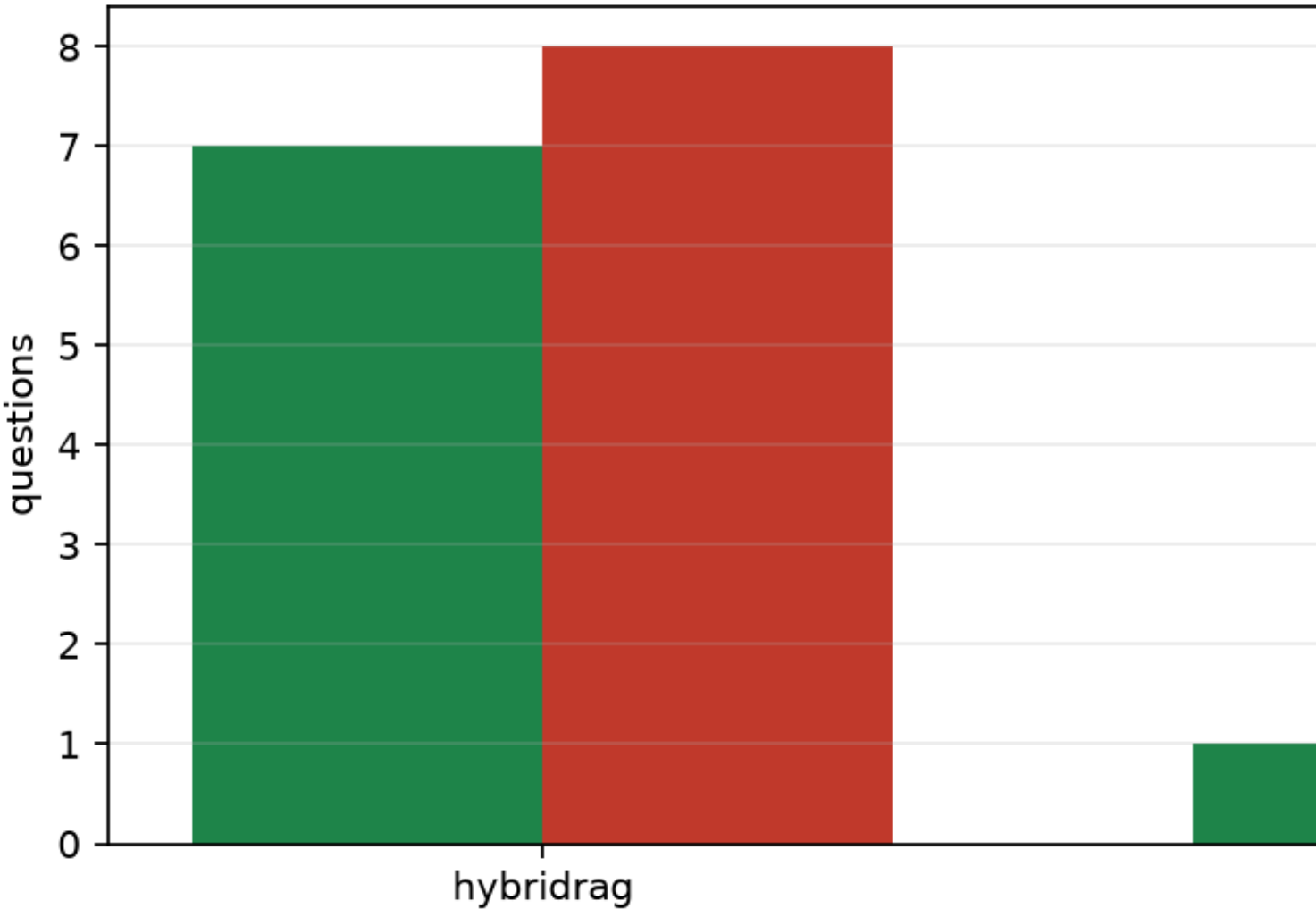
accuracy by strategy

Retrieval strategy vs accuracy (n=54)

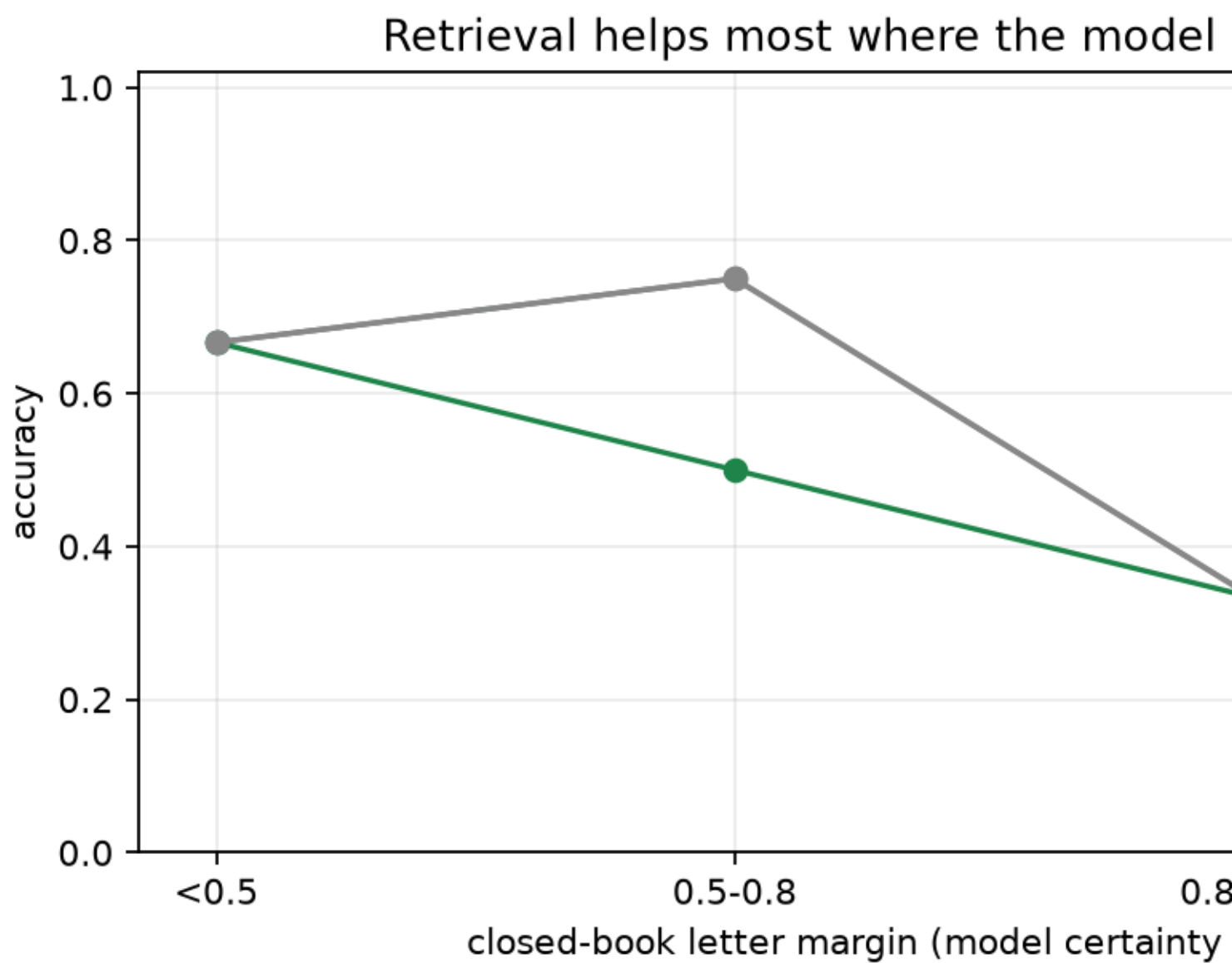


fixed vs broke

Where retrieval changed the a



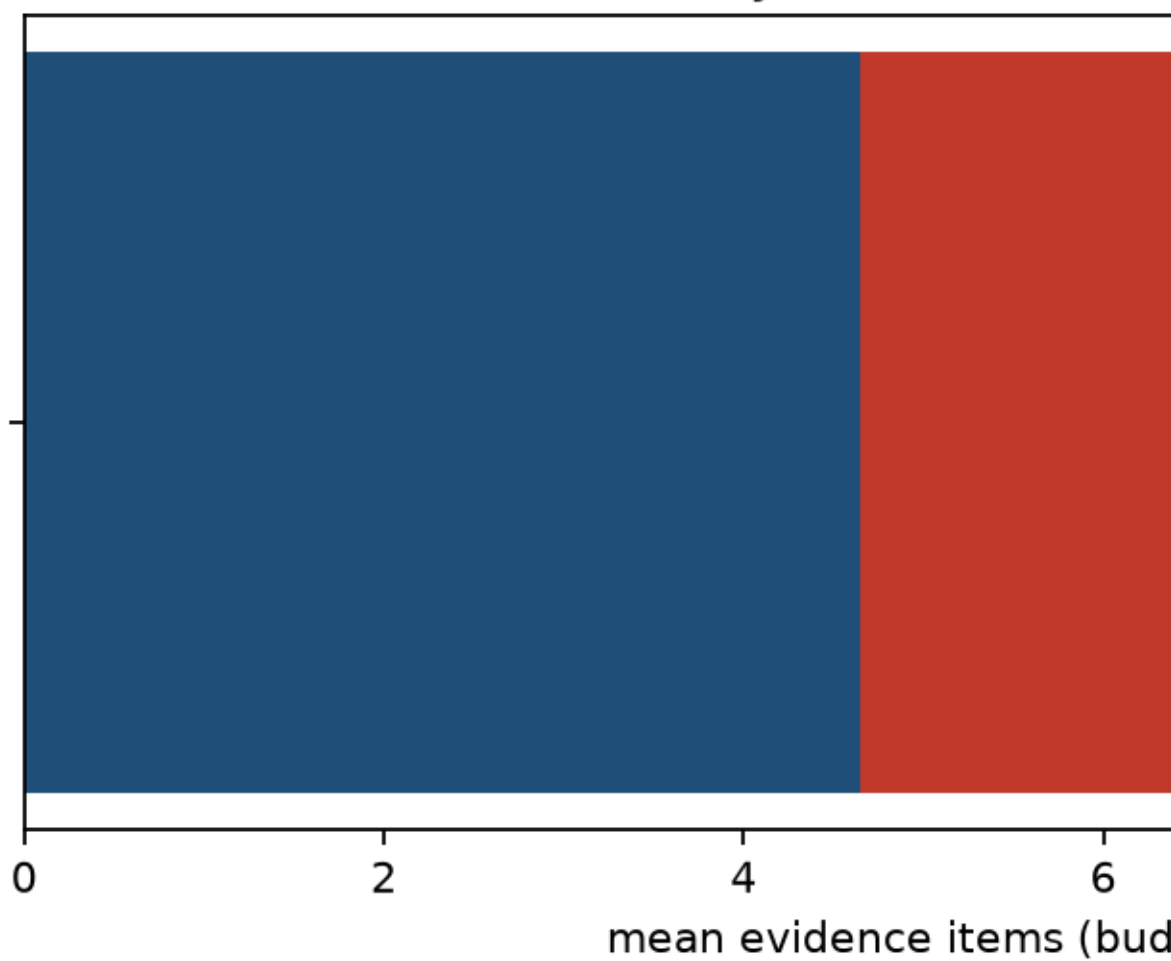
accuracy by margin



hybrid fusion mix

What RRF actually selected into t

fused context



Disagreements (bit order: closed_book, hybridrag, hybridrag_gated)

[101] medqa A 65-year-old man presents to his primary care physician for a change in his b
gold B | close=B hybri=A hybri=B

[101] medqa A 65-year-old man presents to his primary care physician with a neck mass. He
gold B | close=B hybri=D hybri=B

[101] medqa A 72-year-old woman presents to the emergency department with altered mental s
gold A | close=A hybri=D hybri=A

[101] medqa A 68-year-old man presents to the emergency department with confusion. The pat
gold A | close=A hybri=C hybri=A

[101] medqa A 63-year-old man presents to his primary care physician for follow-up. He rep
gold B | close=B hybri=A hybri=B

[010] medqa A 54-year-old man is brought to the emergency department after he was found ou
gold A | close=D hybri=A hybri=D

[010] medqa A 66-year-old man comes to the physician because of fatigue and swelling of hi
gold D | close=C hybri=D hybri=C

[101] medmcqa Most common cause of lobar hemorrhage in elderly age group ?
gold D | close=D hybri=A hybri=D

[011] bioasq Is Semagacestat effective for Alzheimer's Disease?
gold B | close=A hybri=B hybri=B

[010] bioasq Is Lanabecestat effective for Alzheimer's disease?
gold B | close=A hybri=B hybri=A

[101] bioasq Is YKL-40 used as a biomarker for Alzheimer's disease?
gold A | close=A hybri=B hybri=A

[010] bioasq Is Verubecestat effective for Alzheimer's Disease?
gold B | close=A hybri=B hybri=A

[010] bioasq Is verubecestat effective for Alzheimer's Disease?
gold B | close=A hybri=B hybri=A

[100] bioasq Is Semagacestat effective for treatment of Alzheimer's disease?
gold B | close=B hybri=A hybri=A

[010] mmlu A 71-year-old woman is brought to the emergency department by her daughter for
gold D | close=A hybri=D hybri=A